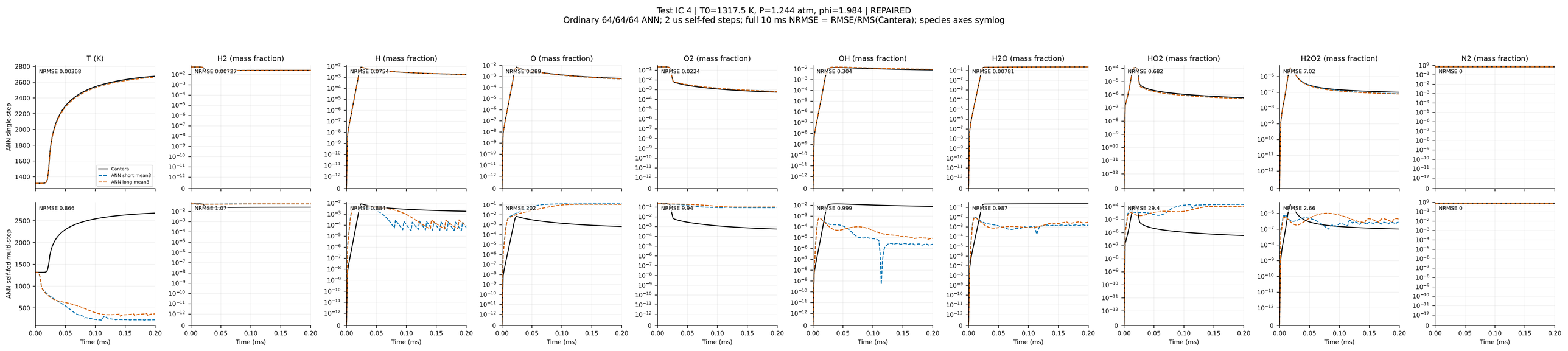
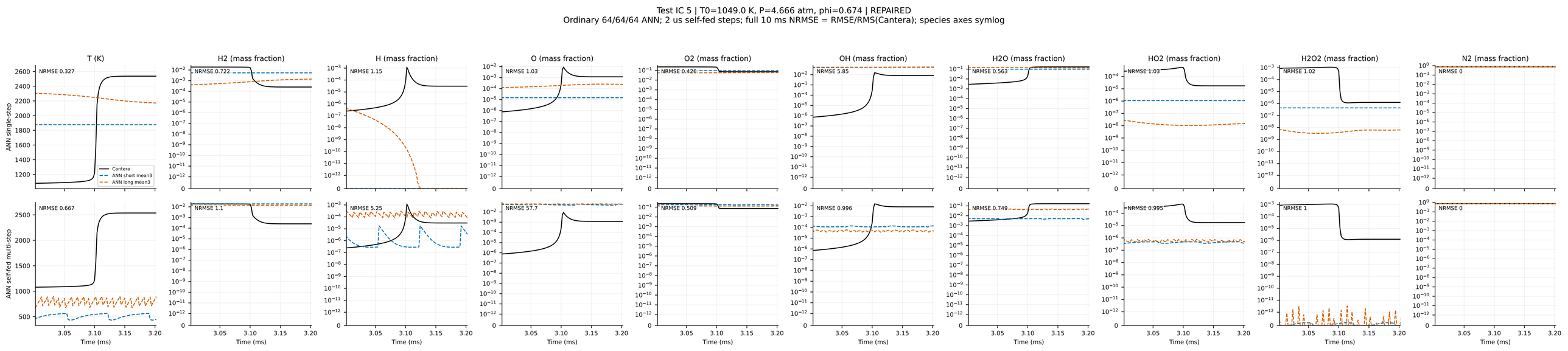
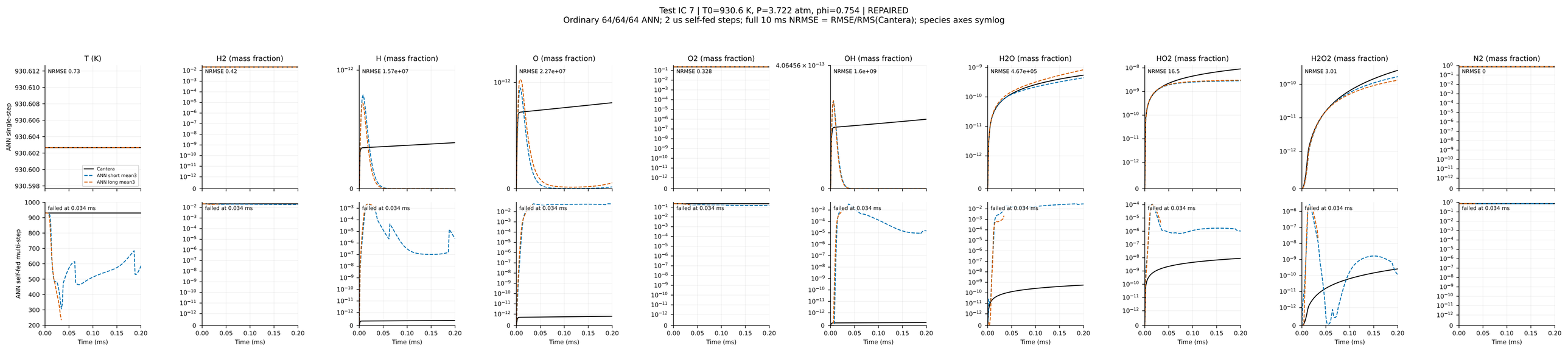
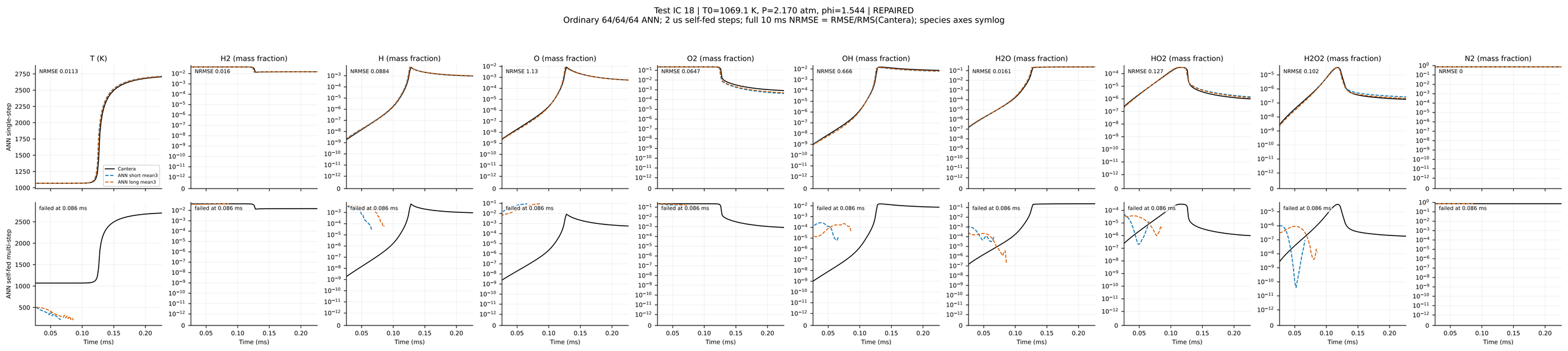
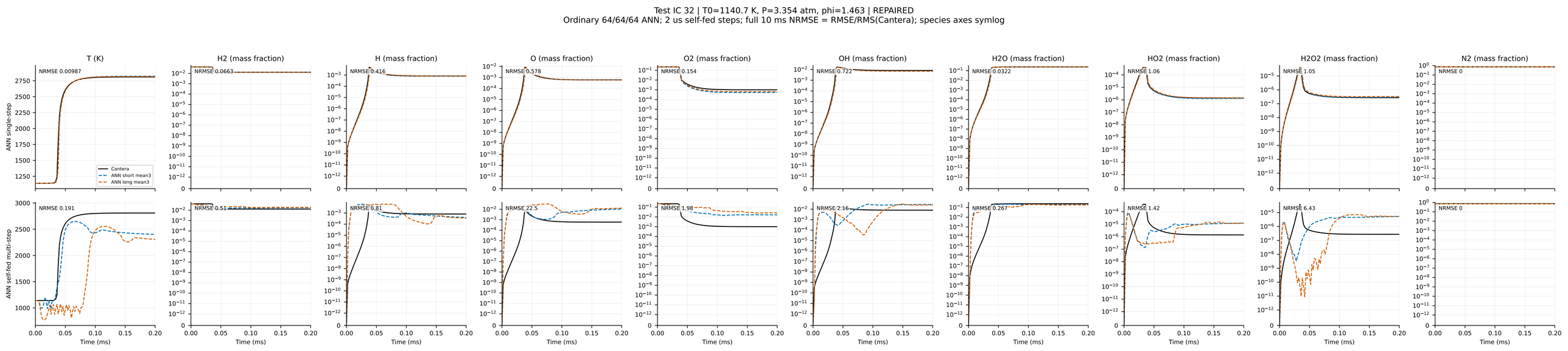
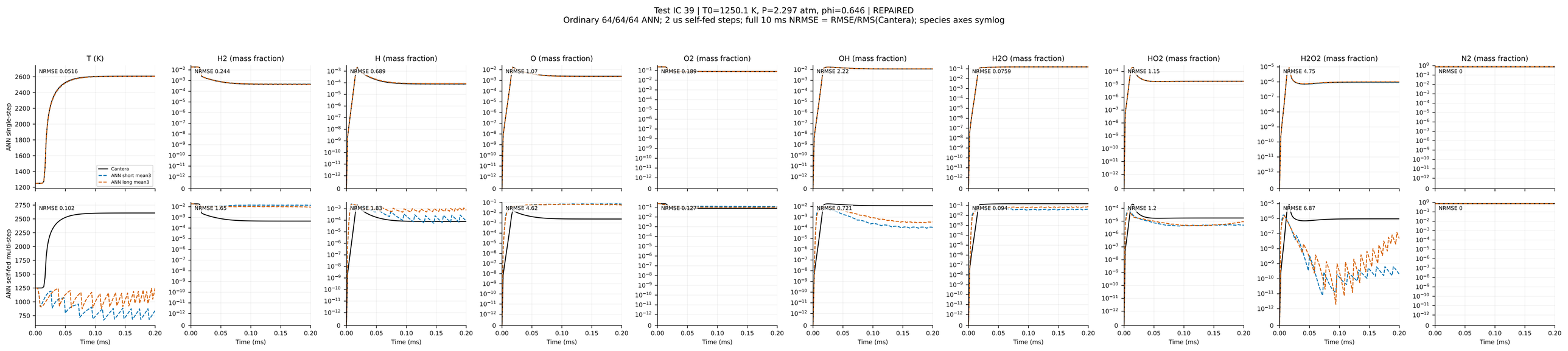

Test IC 4 | T0=1317.5 K, P=1.244 atm, phi=1.984 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.00368
NRMSE 0.00727
NRMSE 0.0754
NRMSE 0.289
NRMSE 0.0224
NRMSE 0.304
NRMSE 0.00781
NRMSE 0.682
NRMSE 7.02
NRMSE 0
NRMSE 0.866
NRMSE 1.07
NRMSE 0.884
NRMSE 202
NRMSE 9.94
NRMSE 0.999
NRMSE 0.987
NRMSE 29.4
NRMSE 2.66
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

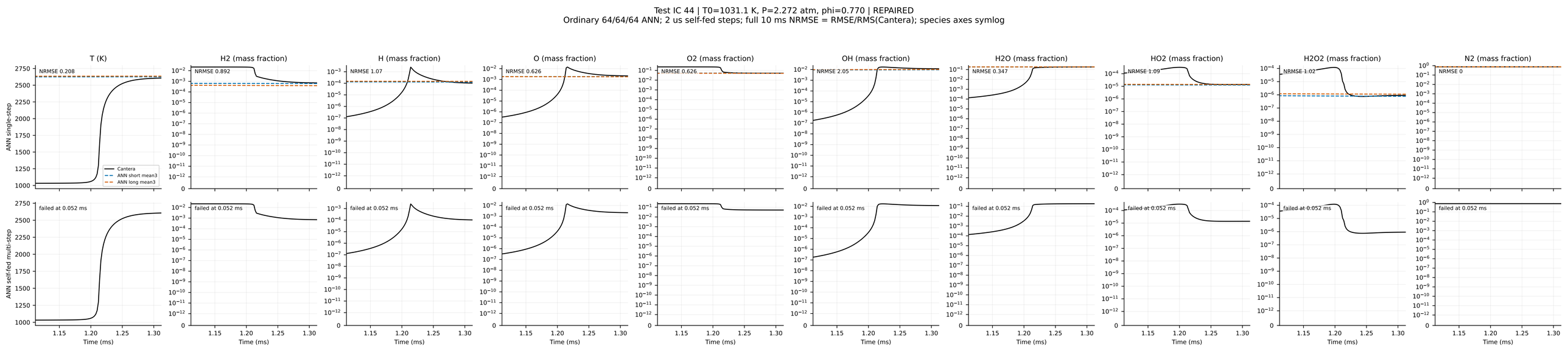

Test IC 5 | T0=1049.0 K, P=4.666 atm, phi=0.674 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.327
NRMSE 0.722
NRMSE 1.15
NRMSE 1.03
NRMSE 0.426
NRMSE 5.85
NRMSE 0.563
NRMSE 1.03
NRMSE 1.02
NRMSE 0
NRMSE 0.667
NRMSE 1.1
NRMSE 5.25
NRMSE 57.7
NRMSE 0.509
NRMSE 0.996
NRMSE 0.749
NRMSE 0.995
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

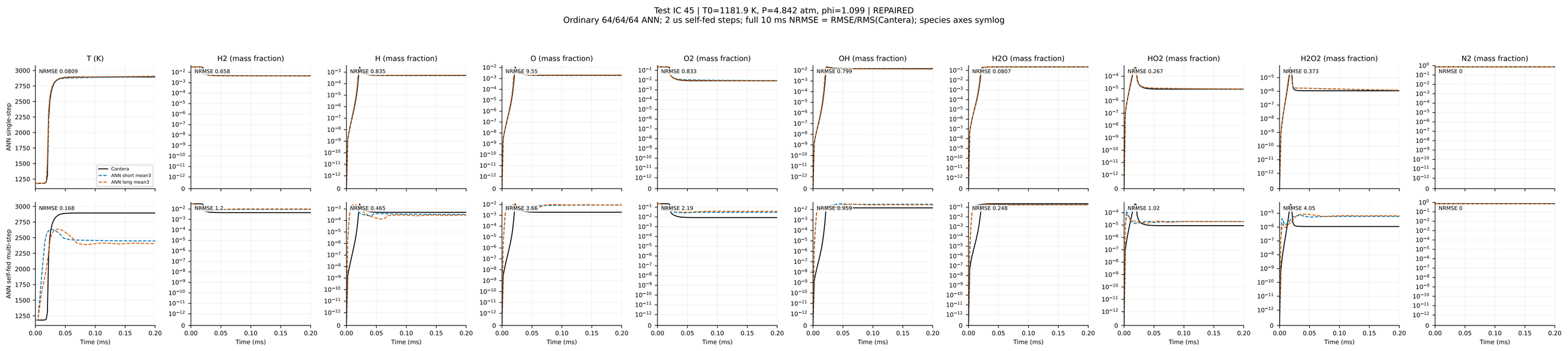

Test IC 7 | T0=930.6 K, P=3.722 atm, phi=0.754 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.73
NRMSE 0.42
NRMSE 1.57e+07
NRMSE 2.27e+07
NRMSE 0.328
NRMSE 1.6e+09
NRMSE 4.67e+05
NRMSE 16.5
NRMSE 3.01
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
failed at 0.034 ms
Time (ms)

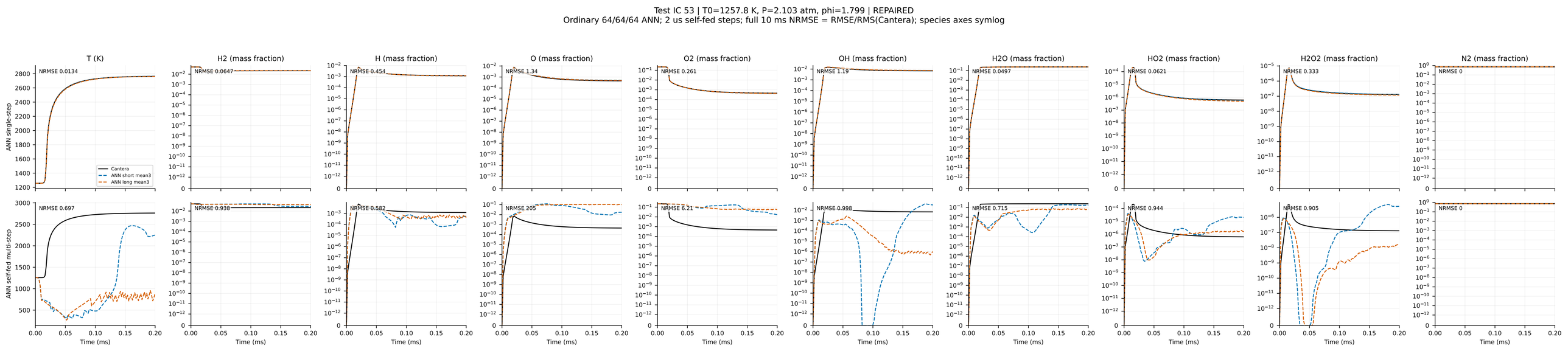

Test IC 18 | T0=1069.1 K, P=2.170 atm, phi=1.544 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0113
NRMSE 0.016
NRMSE 0.0884
NRMSE 1.13
NRMSE 0.0647
NRMSE 0.666
NRMSE 0.0161
NRMSE 0.127
NRMSE 0.102
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
failed at 0.086 ms
Time (ms)

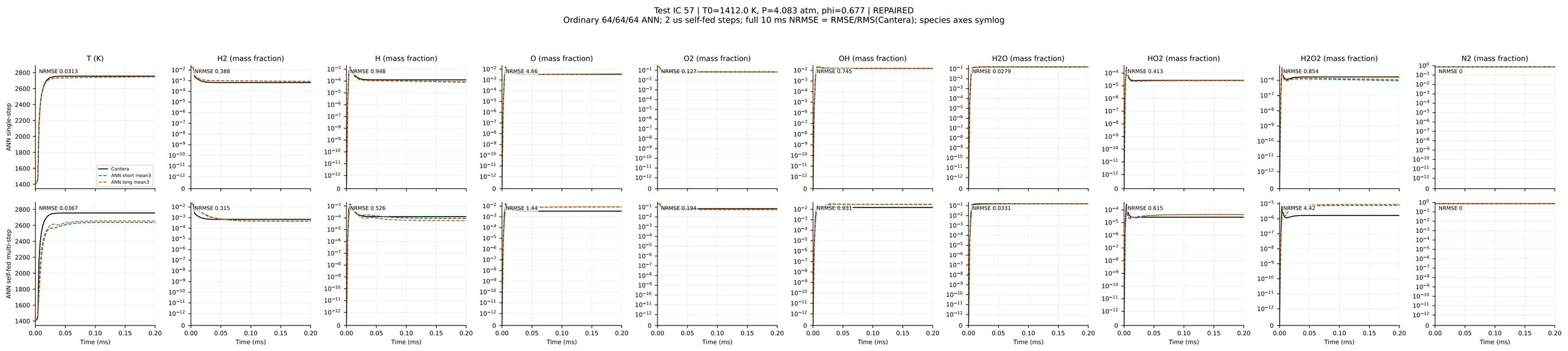

Test IC 32 | T0=1140.7 K, P=3.354 atm, phi=1.463 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.00987
NRMSE 0.0663
NRMSE 0.416
NRMSE 0.578
NRMSE 0.154
NRMSE 0.722
NRMSE 0.0322
NRMSE 1.06
NRMSE 1.05
NRMSE 0
NRMSE 0.191
NRMSE 0.51
NRMSE 0.81
NRMSE 22.5
NRMSE 1.98
NRMSE 2.16
NRMSE 0.267
NRMSE 1.42
NRMSE 6.43
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

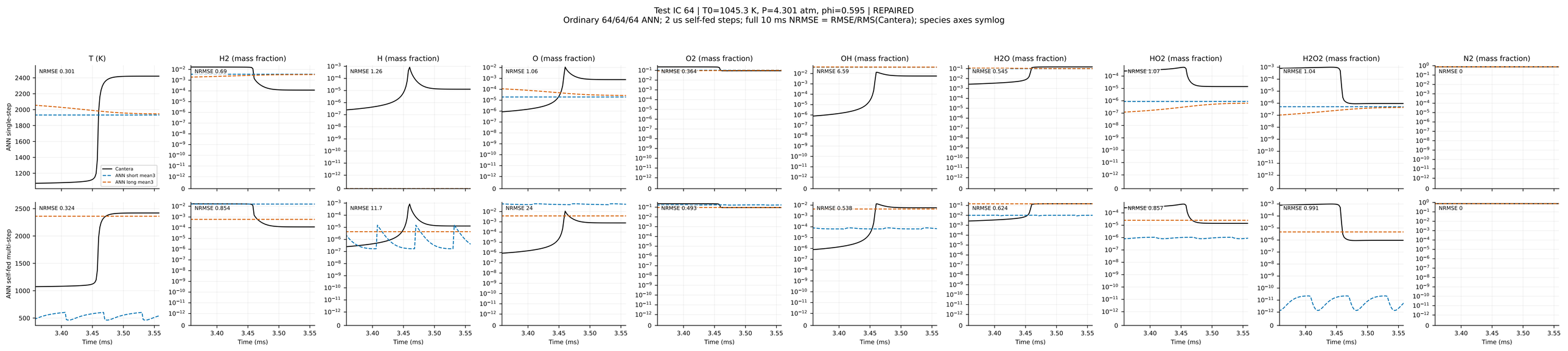

Test IC 39 | T0=1250.1 K, P=2.297 atm, phi=0.646 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
Cantera
ANN short mean3
ANN long mean3
NRMSE 0.0516
NRMSE 0.244
NRMSE 0.689
NRMSE 1.07
NRMSE 0.189
NRMSE 2.22
NRMSE 0.0759
NRMSE 1.15
NRMSE 4.75
NRMSE 0
NRMSE 0.102
NRMSE 1.65
NRMSE 1.83
NRMSE 4.62
NRMSE 0.127
NRMSE 0.721
NRMSE 0.094
NRMSE 1.2
NRMSE 6.87
NRMSE 0
Time (ms)

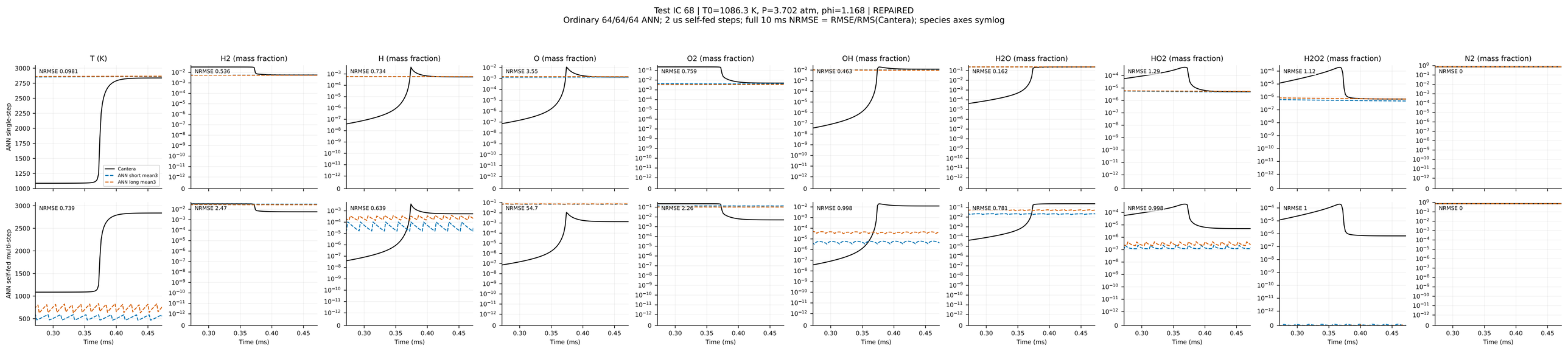

Test IC 44 | T0=1031.1 K, P=2.272 atm, phi=0.770 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.208
NRMSE 0.892
NRMSE 1.07
NRMSE 0.626
NRMSE 0.626
NRMSE 2.05
NRMSE 0.347
NRMSE 1.09
NRMSE 1.02
NRMSE 0
failed at 0.052 ms
Cantera
ANN short mean3
ANN long mean3
Time (ms)

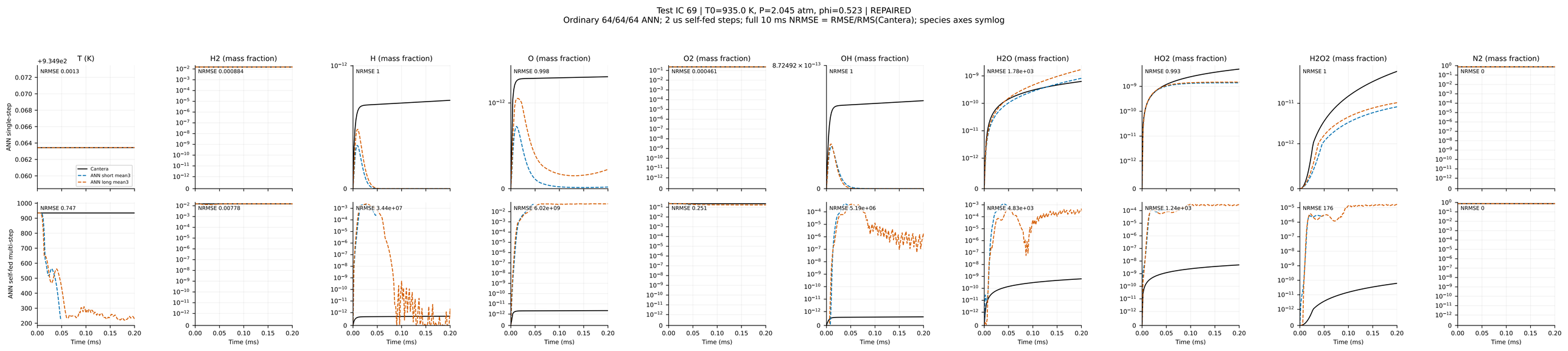

Test IC 45 | T0=1181.9 K, P=4.842 atm, phi=1.099 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0809
NRMSE 0.658
NRMSE 0.835
NRMSE 9.55
NRMSE 0.833
NRMSE 0.799
NRMSE 0.0807
NRMSE 0.267
NRMSE 0.373
NRMSE 0
NRMSE 0.168
NRMSE 1.2
NRMSE 0.465
NRMSE 3.66
NRMSE 2.19
NRMSE 0.959
NRMSE 0.248
NRMSE 1.02
NRMSE 4.05
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

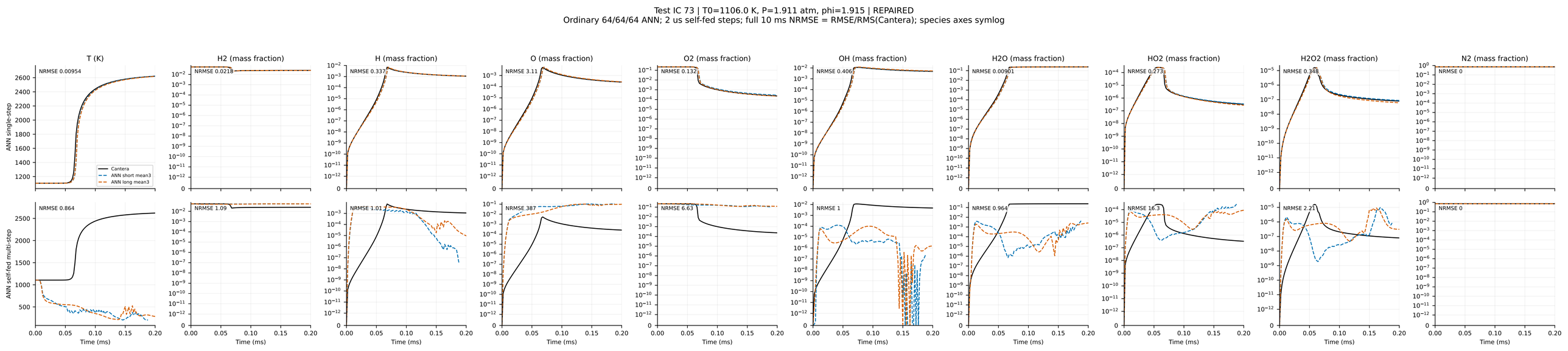

Test IC 53 | T0=1257.8 K, P=2.103 atm, phi=1.799 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0134
NRMSE 0.0647
NRMSE 0.454
NRMSE 1.34
NRMSE 0.261
NRMSE 1.19
NRMSE 0.0497
NRMSE 0.0621
NRMSE 0.333
NRMSE 0
NRMSE 0.697
NRMSE 0.938
NRMSE 0.582
NRMSE 205
NRMSE 6.21
NRMSE 0.998
NRMSE 0.715
NRMSE 0.944
NRMSE 0.905
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

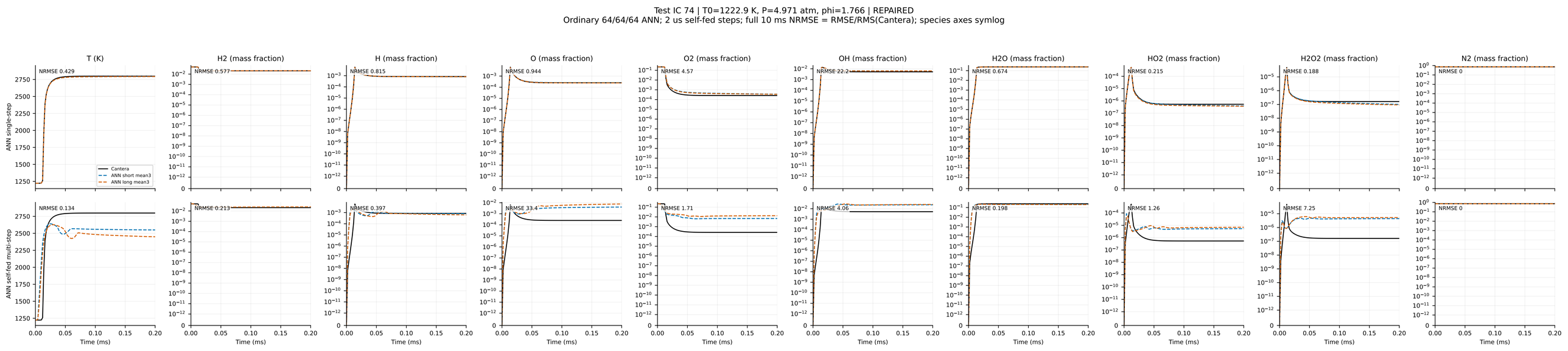

Test IC 57 | T0=1412.0 K, P=4.083 atm, phi=0.677 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0313
NRMSE 0.388
NRMSE 0.948
NRMSE 4.66
NRMSE 0.127
NRMSE 0.745
NRMSE 0.0279
NRMSE 0.413
NRMSE 0.854
NRMSE 0
NRMSE 0.0367
NRMSE 0.315
NRMSE 0.526
NRMSE 1.44
NRMSE 0.194
NRMSE 0.931
NRMSE 0.0331
NRMSE 0.615
NRMSE 4.42
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

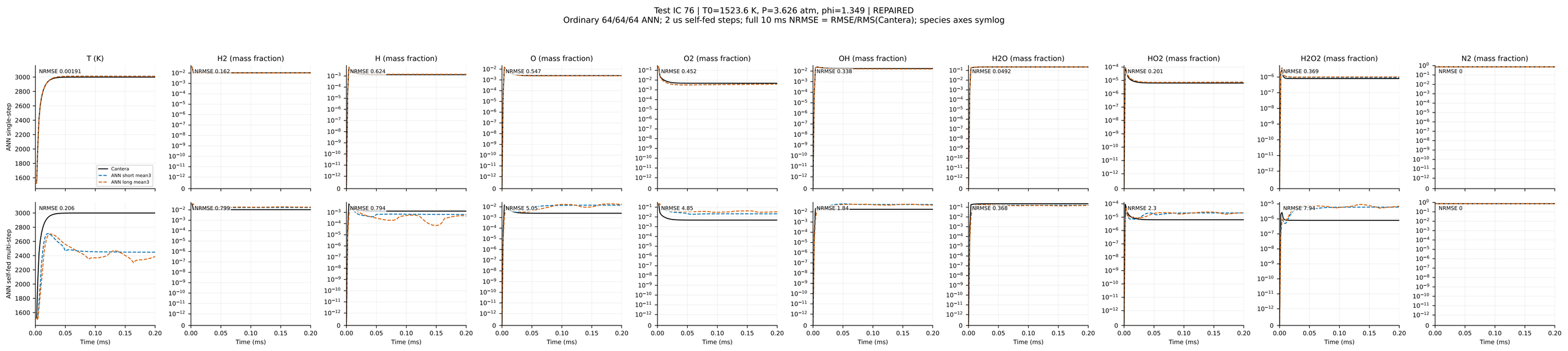

Test IC 64 | T0=1045.3 K, P=4.301 atm, phi=0.595 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.301
NRMSE 0.69
NRMSE 1.26
NRMSE 1.06
NRMSE 0.364
NRMSE 6.59
NRMSE 0.545
NRMSE 1.07
NRMSE 1.04
NRMSE 0
NRMSE 0.324
NRMSE 0.854
NRMSE 11.7
NRMSE 24
NRMSE 0.493
NRMSE 0.538
NRMSE 0.624
NRMSE 0.857
NRMSE 0.991
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

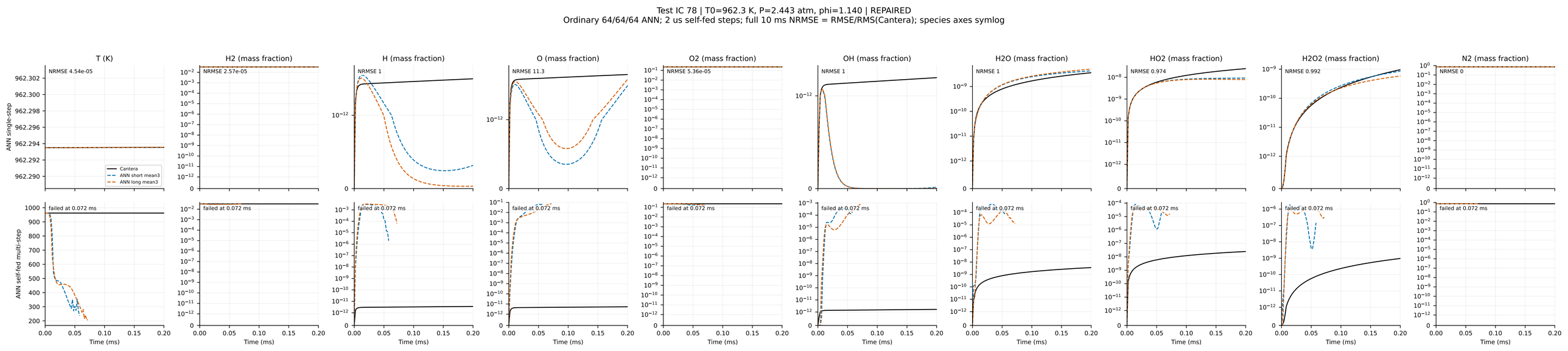

Test IC 68 | T0=1086.3 K, P=3.702 atm, phi=1.168 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0981
NRMSE 0.536
NRMSE 0.734
NRMSE 3.55
NRMSE 0.759
NRMSE 0.463
NRMSE 0.162
NRMSE 1.29
NRMSE 1.12
NRMSE 0
NRMSE 0.739
NRMSE 2.47
NRMSE 0.639
NRMSE 54.7
NRMSE 2.26
NRMSE 0.998
NRMSE 0.781
NRMSE 0.998
NRMSE 1
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)

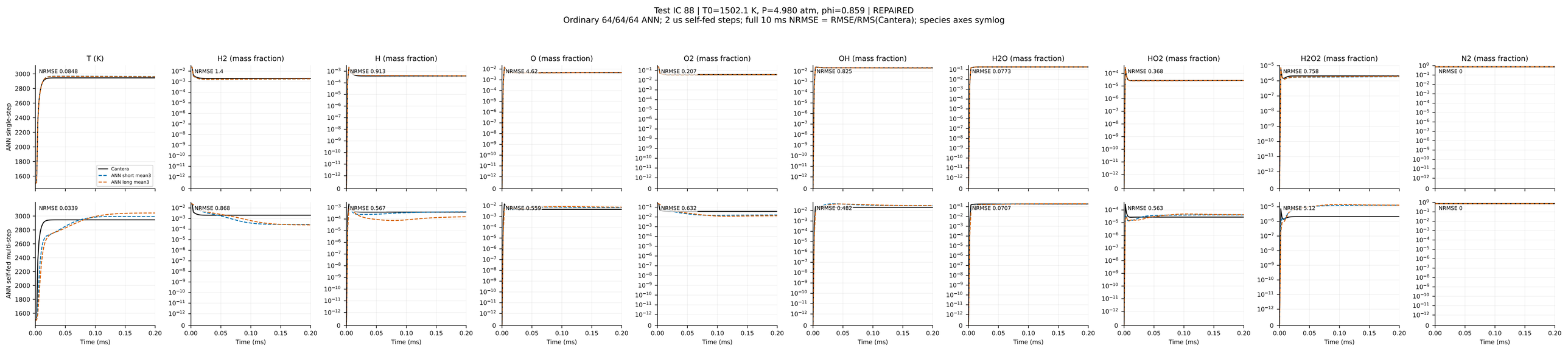

Test IC 69 | T0=935.0 K, P=2.045 atm, phi=0.523 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
+9.349e2
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
8.72492 × 10−13
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0013
NRMSE 0.000884
NRMSE 1
NRMSE 0.998
NRMSE 0.000461
NRMSE 1
NRMSE 1.78e+03
NRMSE 0.993
NRMSE 1
NRMSE 0
NRMSE 0.747
NRMSE 0.00778
NRMSE 3.44e+07
NRMSE 6.02e+09
NRMSE 0.251
NRMSE 5.19e+06
NRMSE 4.83e+03
NRMSE 1.24e+03
NRMSE 176
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)


Test IC 73 | T0=1106.0 K, P=1.911 atm, phi=1.915 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.00954
NRMSE 0.0218
NRMSE 0.337
NRMSE 3.11
NRMSE 0.132
NRMSE 0.406
NRMSE 0.00901
NRMSE 0.273
NRMSE 0.348
NRMSE 0
NRMSE 0.864
NRMSE 1.09
NRMSE 1.01
NRMSE 387
NRMSE 6.63
NRMSE 1
NRMSE 0.964
NRMSE 16.3
NRMSE 2.21
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)


Test IC 74 | T0=1222.9 K, P=4.971 atm, phi=1.766 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.429
NRMSE 0.577
NRMSE 0.815
NRMSE 0.944
NRMSE 4.57
NRMSE 22.2
NRMSE 0.674
NRMSE 0.215
NRMSE 0.188
NRMSE 0
NRMSE 0.134
NRMSE 0.213
NRMSE 0.397
NRMSE 33.4
NRMSE 1.71
NRMSE 4.06
NRMSE 0.198
NRMSE 1.26
NRMSE 7.25
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)


Test IC 76 | T0=1523.6 K, P=3.626 atm, phi=1.349 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.00191
NRMSE 0.162
NRMSE 0.624
NRMSE 0.547
NRMSE 0.452
NRMSE 0.338
NRMSE 0.0492
NRMSE 0.201
NRMSE 0.369
NRMSE 0
NRMSE 0.206
NRMSE 0.799
NRMSE 0.794
NRMSE 5.05
NRMSE 4.85
NRMSE 1.84
NRMSE 0.368
NRMSE 2.3
NRMSE 7.94
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)


Test IC 78 | T0=962.3 K, P=2.443 atm, phi=1.140 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 4.54e-05
NRMSE 2.57e-05
NRMSE 1
NRMSE 11.3
NRMSE 5.36e-05
NRMSE 1
NRMSE 1
NRMSE 0.974
NRMSE 0.992
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
failed at 0.072 ms
Time (ms)


Test IC 88 | T0=1502.1 K, P=4.980 atm, phi=0.859 | REPAIRED
Ordinary 64/64/64 ANN; 2 us self-fed steps; full 10 ms NRMSE = RMSE/RMS(Cantera); species axes symlog
T (K)
H2 (mass fraction)
H (mass fraction)
O (mass fraction)
O2 (mass fraction)
OH (mass fraction)
H2O (mass fraction)
HO2 (mass fraction)
H2O2 (mass fraction)
N2 (mass fraction)
ANN single-step
ANN self-fed multi-step
NRMSE 0.0848
NRMSE 1.4
NRMSE 0.913
NRMSE 4.62
NRMSE 0.207
NRMSE 0.825
NRMSE 0.0773
NRMSE 0.368
NRMSE 0.758
NRMSE 0
NRMSE 0.0339
NRMSE 0.868
NRMSE 0.567
NRMSE 0.559
NRMSE 0.632
NRMSE 0.482
NRMSE 0.0707
NRMSE 0.563
NRMSE 5.12
NRMSE 0
Cantera
ANN short mean3
ANN long mean3
Time (ms)